

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) April 2, 2011

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1: Consider the following two Latin square designs of order 4 and answer Questions A, B and C

Design (I)					Design (II)				
Column					Column				
Row	1	2	3	4	Row	1	2	3	4
1	A	B	C	D	1	A	C	B	D
2	D	A	B	C	2	D	B	C	A
3	C	D	A	B	3	C	A	D	B
4	B	C	D	A	4	B	D	A	C

A. Do you see any potential shortcomings for using design (I)?

Now Consider Latin square design (II) up above and the data as follows.

Column				
Row	1	2	3	4
1	235	236	218	268
2	251	241	227	229
3	234	273	274	226
4	195	270	230	225

Recall that the linear model for this Latin square design is

$$y_{ijl} = \eta + \alpha_i + \beta_j + \tau_l + \varepsilon_{ijl}, \quad i = 1, \dots, 4; j = 1, \dots, 4; l = 1, \dots, 4,$$

Where l = Latin letter in the (i, j) cell of the Latin square, $\alpha_i = i$ throw effect, $\beta_j = j$ the column effect, $\tau_l = l$ th treatment (i.e., Latin letter) effect, ε_{ijl} are independent

$N(0, \sigma^2)$. Assume the zero-sum constraints $\sum_{i=1}^4 \alpha_i = \sum_{j=1}^4 \beta_j = \sum_{l=1}^4 \tau_l = 0$.

B. Calculate $\hat{\alpha}_3$, $\hat{\beta}_1$, $\hat{\tau}_2$, you can use the fact $\bar{y}_{...} = 239.5$.

C. What is the error/residual degrees of freedom for this design?

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- 2 Assume that the Distributions of profits, P in a year per acre for Wheat farms in Minnesota are distributed $N(u, \sigma)$ with $u = 1000 X$ and $\sigma = 300 X$ where X is the number of inches of rain per year that acre received. *NOTE – For these questions do not worry about the size of the farm you can assume that all farms are 1 acre in size*

A. If $X \sim U[10, 20]$ give the overall expectation and variance of P in a one acre farm

B. Let A and B be two 1 acre farms in Minnesota. Let $X_A = 15$ for farm A and $X_B = 20$.

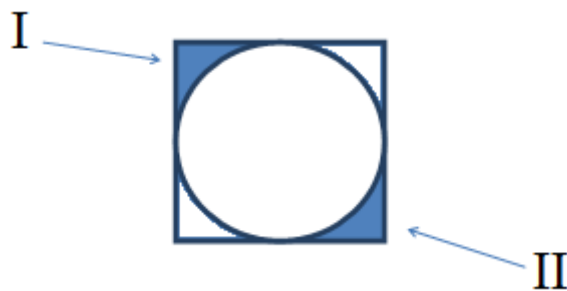
What is the Distribution of $P_A - P_B$ if

- P_A and P_B are independent
- $\text{Corr } P_A, P_B$ is 1

3: There are three coins in a barrel. These coins, when flipped, will come up heads with respective probabilities 0.4, 0.5, and 0.9. A coin is randomly selected from among these three and is then flipped ten times

- a. What is the probability of having 0 heads?
- b. Let N be the number of heads obtained on the ten flips. Find $P(N=n)$, where $n=0,1,\dots,10$.
- c. If you can win one dollar each time a head appears, how much money should you lose each time a tail appears in order to have an expected gain of 0?
- d. Now the rule is changed. We select one coin each time at random and continue this process for 10 times with replacement. What is $P(N=n)$?

4. A rocket company is firing missile into a square region with each side of length of $2r$ that has an inscribed circle with a radius r as shown below.



One of the scientists believes that the rockets are biased to go towards the upper left and away from the lower right. Specifically that the rocket is more likely to land in Region I than it is to land in Region II.

- A. State the scientist's Null and Alternative Hypothesis and all the assumptions needed to test these.
- B. What are the expected numbers of rockets each that land in Region I and in Region II if H_0 is true
- C. Out of 1,000 shots, fired, 63 have landed in Region I and 44 have landed in Region II. Give a best large sample statistical test of the Null Hypothesis against the Alternative at $\alpha = 0.05$